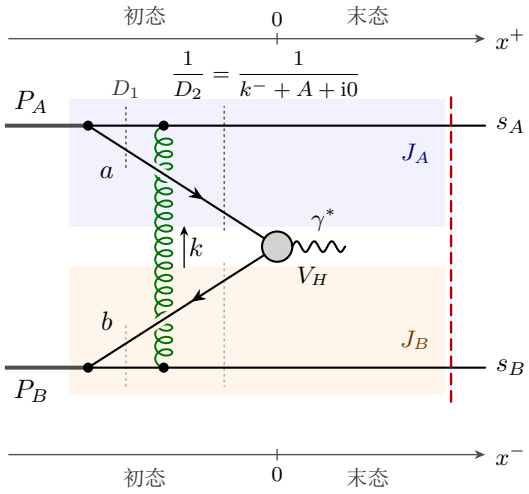
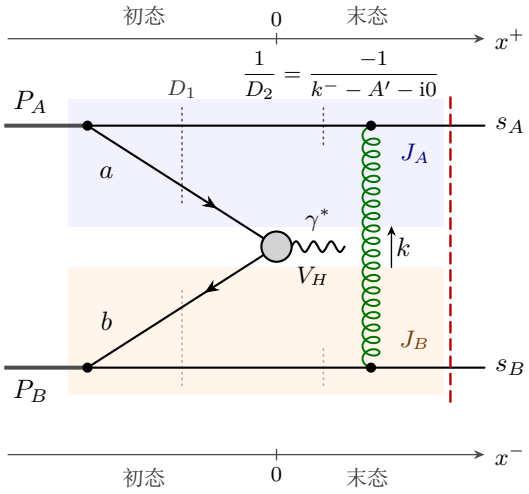
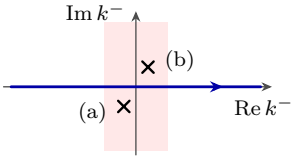


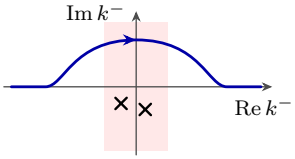
(a) k 在 V_H 之前附着：只进入初态



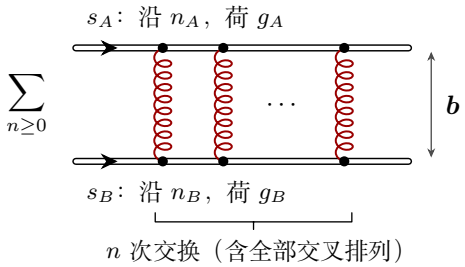
(b) k 在 V_H 之后附着：只进入末态



(c) (a)+(b)：极点分居两侧，k^- 围道被钳住



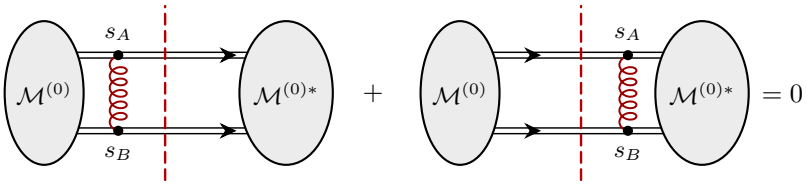
(d) 只经过初态：极点同侧，k^- 围道可移入上半平面



$$= \sum_{n\geq 0} \frac{\big(\mathrm{i}\chi(\boldsymbol{b})\big)^n}{n!} = \mathrm{e}^{\mathrm{i}\chi(\boldsymbol{b})}$$

$$\chi(\boldsymbol{b}) = -\frac{g_A g_B}{2\pi} K_0(m|\boldsymbol{b}|)$$

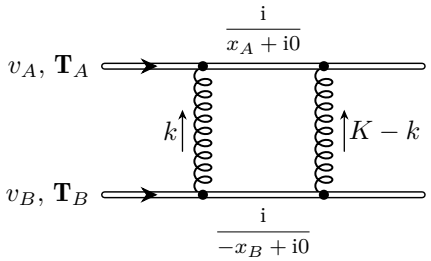
(a) 程函旁观者之间 n 次有质量阿贝尔玻色子交换之和为相位



$+\mathrm{i}\chi(\boldsymbol{b})\,|\tilde{\mathcal{M}}^{(0)}(\boldsymbol{b})|^2$ $\qquad$ $-\mathrm{i}\chi(\boldsymbol{b})\,|\tilde{\mathcal{M}}^{(0)}(\boldsymbol{b})|^2$

$\mathcal{O}\big(g^{2n}\big)$：$\displaystyle\sum_{j=0}^{n} \frac{(\mathrm{i}\chi)^j}{j!}\,\frac{(-\mathrm{i}\chi)^{n-j}}{(n-j)!} = 0$ $(n \geq 1)$；全阶：$|\mathrm{e}^{\mathrm{i}\chi(\boldsymbol{b})}|^2 = 1$

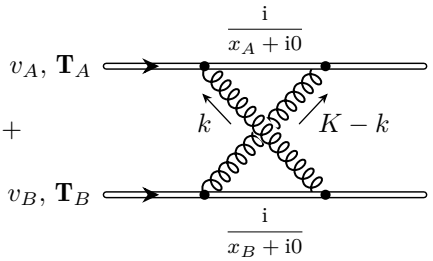
(b) 割线位置之和（对旁观者相对横动量 $\boldsymbol{\Delta}$ 积分，即在 $\boldsymbol{b}$ 空间中）



$$J_{\mathrm{box}} = \frac{2\pi}{ab}\,(\pi + \mathrm{i}Y)$$

$$c_{\mathrm{box}} = X^2$$

(a) 非交叉图（box）

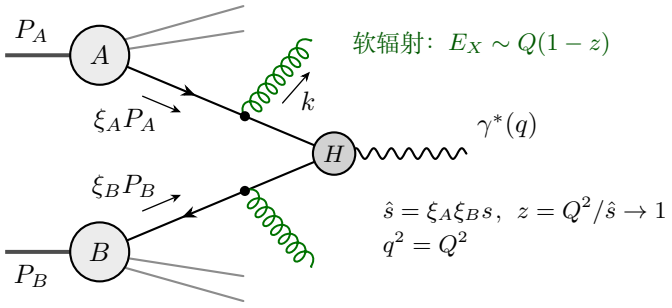


$$J_{\mathrm{cross}} = -\frac{2\pi\mathrm{i}Y}{ab}$$

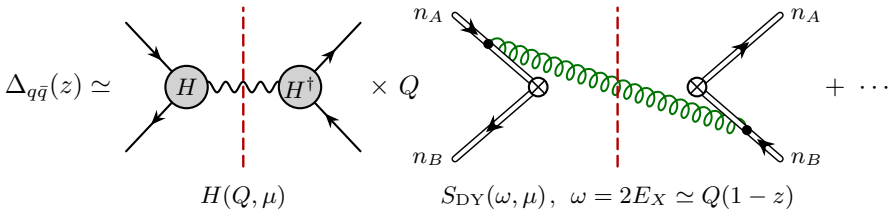
$$c_{\mathrm{cross}} = X^2 + \frac{C_A}{2}\,X$$

(b) 交叉图（crossed box）

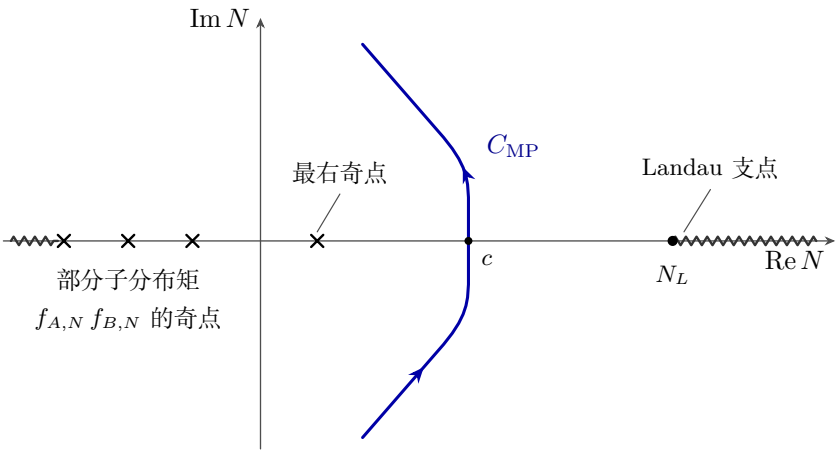
$$J_{\mathrm{box}}\,c_{\mathrm{box}} + J_{\mathrm{cross}}\,c_{\mathrm{cross}} = \frac{2\pi}{ab}\Big[\underbrace{\pi X^2}_{\text{指数化项}}\ \underbrace{-\mathrm{i}Y\,\tfrac{C_A}{2}\,X}_{\text{Regge 相位}}\Big]$$



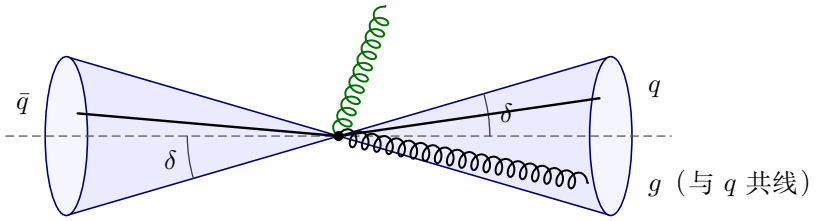
(a) 部分子阈值：末态只有软辐射



(b) 领头幂次下硬系数的重新因子化（“$\cdots$”为 $(1-z)$ 的次领头幂次项）

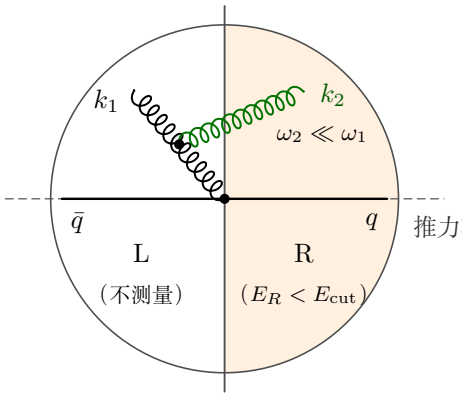

Im N
C_{MP}
最右奇点
Landau 支点
c
N_L
Re N
部分子分布矩
$f_{A,N}$ $f_{B,N}$ 的奇点

锥外的软胶子：$E < \epsilon\sqrt{s}$



两喷注事例：两锥之外的总能量 $< \epsilon\sqrt{s}$

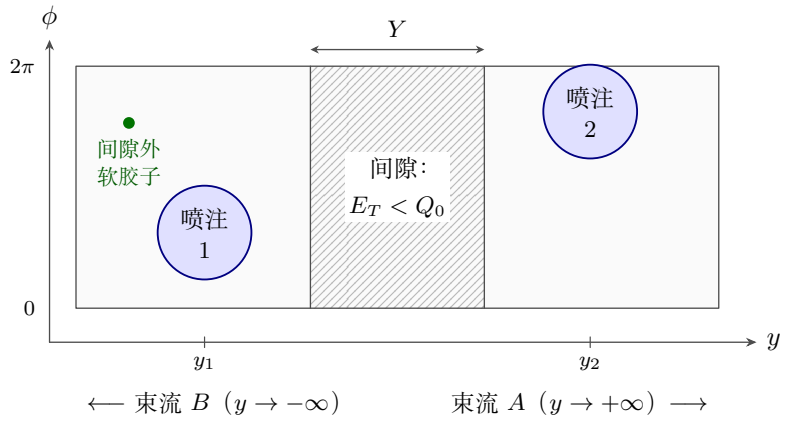
（能量份额 ϵ 相对于总能量 $\sqrt{s}$，不是束流能量 $\sqrt{s}/2$；$\epsilon < \delta \ll 1$）



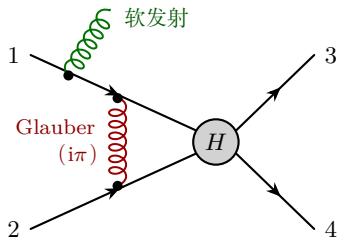
$\boxed{\mathcal{O}(\alpha_s^2)\text{：}\ C_F C_A}$

$k_1 \in \text{L}$：不受限制

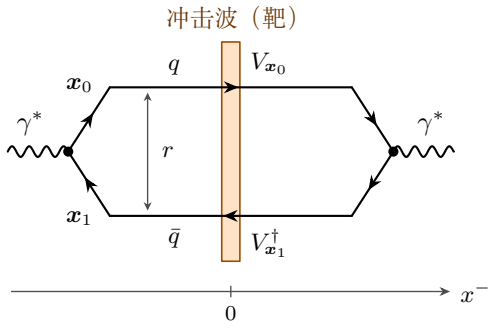
$k_2 \in \text{R}$：受否决



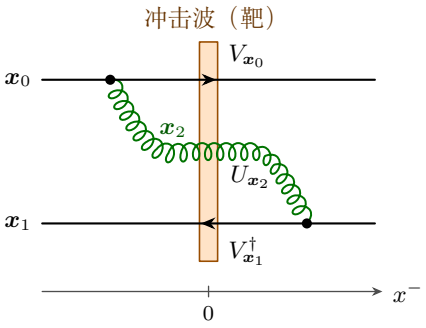
(a) (y, ϕ) 平面中的间隙可观测量



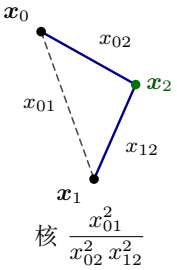
(b) 初态 Glauber 相位与软发射



(a) $q\bar{q}$ 偶极子穿过冲击波



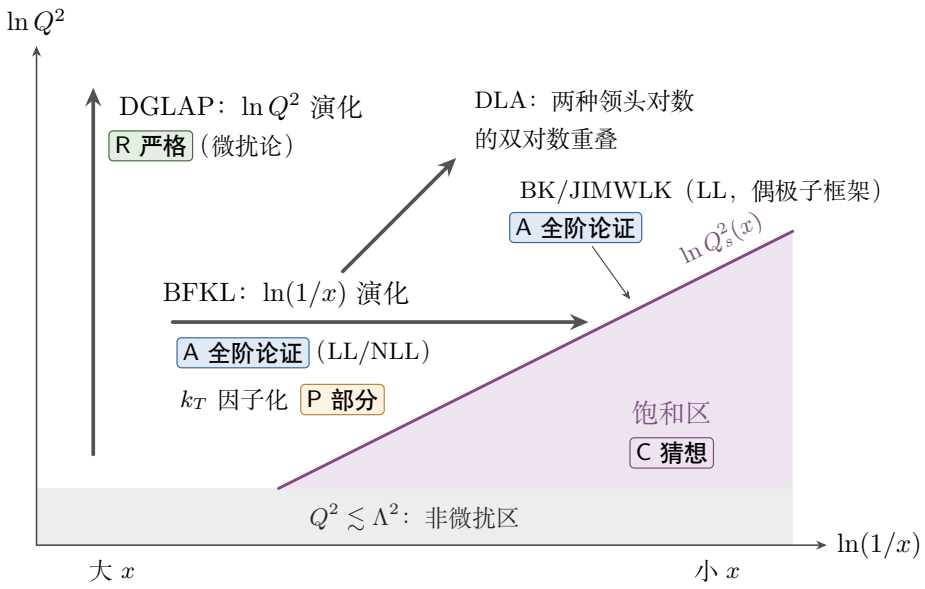
(b) 软胶子在 $\boldsymbol{x}_2$ 穿过冲击波



(c) 横向平面

$$S_{01} = \frac{1}{N_c}\,\mathrm{Tr}\left[V_{\boldsymbol{x}_0} V^\dagger_{\boldsymbol{x}_1}\right], \qquad N_{01} = 1 - S_{01}\,, \qquad \bar{\alpha}_s = \frac{N_c \alpha_s}{\pi}$$

$$\partial_Y N_{01} = \frac{\bar{\alpha}_s}{2\pi} \int \mathrm{d}^2 x_2\, \frac{x_{01}^2}{x_{02}^2 x_{12}^2} \left[N_{02} + N_{12} - N_{01} - N_{02} N_{12}\right]$$


$\ln Q^2$
DGLAP：$\ln Q^2$ 演化
R 严格（微扰论）
DLA：两种领头对数
的双对数重叠
BK/JIMWLK（LL，偶极子框架）
A 全阶论证
$\ln Q_s^2(x)$
BFKL：$\ln(1/x)$ 演化
A 全阶论证（LL/NLL）
k_T 因子化 P 部分
饱和区
C 猜想
$Q^2 \lesssim \Lambda^2$：非微扰区
$\ln(1/x)$
大 x
小 x